

Assignment 5: Developing the Software Requirements Specification (SRS)

Group# 19

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Introduction:

▪ Purpose:

- The Pakistan Railways Website solves the core problem of lack of coordination between Pakistan Railways departments by letting authorized employees access their departments from anywhere in the world and allowing online communication between the departments. Hence, it fixes the main problem of “delay in train arrivals and departures.”
- The intended audience of SRS includes
 - Ministry of Railways Pakistan
 - All authorized users of the system
 - Software development team

Scope:

Scope Elements:

Product Functionality:

Key Features And Functionalities:

- Online coordination between Railway departments
- Real-time updates for both passenger and employee sides.
- Secure access of user data to only authorized personnel
- Web-based from anywhere, for both passengers and employees

Benefits:

- Proper coordination between departments.
- Proper functioning of departments.
- Satisfaction of customers.
- Healthy generation of revenue.
- No delays in train timings.

Objectives:

- Allow online coordination of departments of Railways

Project Resources:

- **Team Size:**
 - 14 people
- **Skill Set:**
 - 1 Project Owner (Stakeholder)
 - 1 Project Manager
 - 3 Software Developers
 - 2 UI designers
 - 2 Quality assurance engineers
 - 3 Requirement engineers
 - 1 System architect
 - 1 Scrum master
- **Tools available:**
 - **IDE:** VS Code (HTML, CSS & Javascript, Node.js, React.js)
 - **Database:** MongoDB
 - **Task Management:** Scrumwise
 - **Requirements Management:** Pivotal Tracker
 - **Software Life Cycle Management:** Code beamer ALM

Available Time:**→ Total Time: 5 Months:**

- ◆ **Requirements analysis:** 2 Months
- ◆ **System design:** 1 Month
- ◆ **Implementation:** 1 Months
- ◆ **Testing:** 29 Days
- ◆ **Deployment:** 1 Day

▪ Definitions, acronyms, and abbreviations

- **IEEE** - Institute of Electrical and Electronics Engineers
- **HTTPS** - HyperText Transfer Protocol Secure
- **PC** - Personal Computer
- **NFR** - Non-Functional Requirement

▪ References

- SRS format is based on IEEE 830-1998 Specification

▪ Overview

The rest of this document covers

- Diagrammatic description
- User classes
- Assumptions and dependencies
- Focus on requirements

The organization of this document is as follows:

- Introduction
- Overall Description
- Specific Requirements
- Use Cases
- Appendices

Overall Description

▪ Product perspective

Customer perspective of our software

Utility: Users can

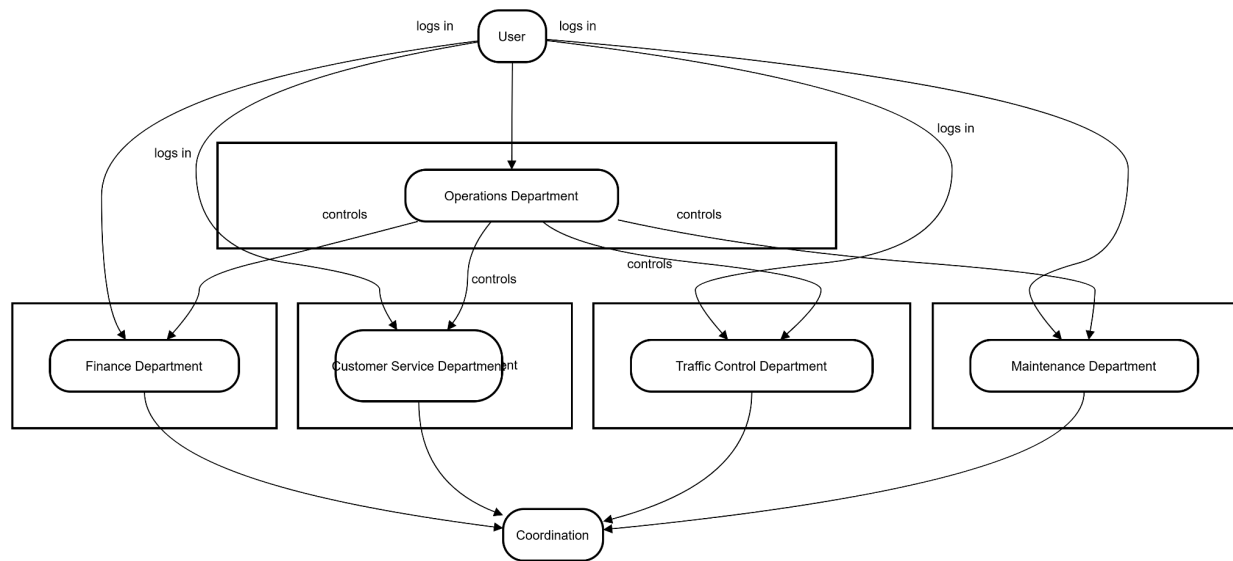
- Log in to their respective employee accounts
- Use their department to interact with other departments
- Have control over the functionality of their department

Ease of Use:

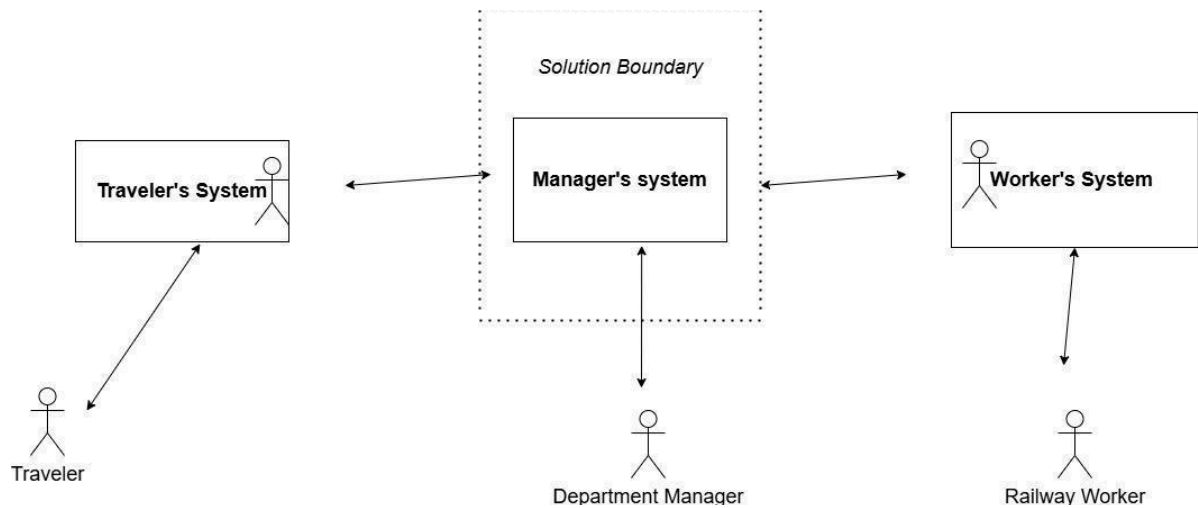
- Pakistan Railways Website is accessible from any modern PC and mobile phone
- The website is accessible from anywhere in the world
- The accounts can be easily logged in and logged out of

Dependency:

Pakistan Railways Website is not dependent on any other software, it is a self-contained website.

Overall Arcitechture:

Scope diagram



▪ Product Features

Customer Benefit	Supporting Features
Reduced Delays	Improved departmental coordination and real-time updates
Real-Time Monitoring	Live tracking of train engine locations and status
Secure Access	User data accessible only to authorized personnel
Ease of Use	Web-based platform accessible by railway staff from any location

▪ User classes and characteristics

1. User Class

Attributes:

userID: int: A unique identifier for each user, likely for authentication and tracking.

name: String: Represents the name of the user.

role: String: Indicates the user's role in the system (e.g., administrator, operator).

Methods:

login(): Allows the user to authenticate and gain access to the system.

requestData(): Enables the user to request specific information from the system.

Analysis:

The User class represents the entry point for interaction with the system.

Through roles, users are granted permissions to interact with specific departments or functionalities.

Likely designed to support multiple user types with varying levels of system access.

2. System Class

Methods:

accessDepartment(): Facilitates user access to different departments within the system.

Analysis:

Acts as the central coordinator of the software, managing how users and departments interact.

The accessDepartment() method indicates role-based or condition-based routing of requests.

3. FinanceDepartment Class

Methods:

acceptReport(): Accepts financial reports submitted by users or other departments.

rejectReport(): Rejects reports that do not meet specified criteria.

Analysis

Focuses on report handling, specifically financial data.

Likely interacts with OperationsDepartment or User for report submission and approval workflows.

4. TrafficControlDepartment Class

Methods:

requestDetailsOfTrain(Train obj): Retrieves detailed information about a specific train.

signalEmergency(Train obj): Handles emergency situations related to a train.

Analysis:

Manages the system's operational safety by monitoring and controlling train activities.

Plays a critical role in ensuring safety through emergency signaling and train monitoring.

5. CustomerServiceDepartment Class

Methods:

checkMessages(): Allows customer service to view messages or inquiries.

sendMessages(): Enables sending responses or notifications to users.

Analysis:

Focuses on user interaction and communication.

Likely integrates with external systems or user interfaces for real-time messaging.

6. ResourceManagementDepartment Class

Methods:

showCurrentResources(): Displays the current status of resources like fuel, materials, etc.

orderMoreResources(): Facilitates resource replenishment when levels are low.

Analysis:

Handles resource tracking and procurement for the system.

Plays a supportive role in ensuring operational continuity, particularly in relation to the Storage class.

7. OperationsDepartment Class

Methods:

reportProblem(): Allows users to report issues or operational problems.

requestPerformanceReport(): Enables requesting performance data for operational analysis.

Analysis:

Acts as the system's monitoring and reporting hub for operations.

Interacts with other departments like MaintenanceDepartment to resolve issues and ensure efficiency.

8. MaintenanceDepartment Class

Methods:

acceptReport(): Approves and processes maintenance reports.

rejectReport(): Rejects invalid or unnecessary maintenance reports.

Analysis:

Specializes in handling system maintenance-related issues.

Likely interacts with OperationsDepartment and Train objects for resolving issues or scheduling maintenance.

9. Train Class

Attributes:

modelNo: String: Identifies the train model.

departure: String: Represents the departure location.

arrival: String: Represents the arrival location.

Analysis:

Represents a train object in the system, encapsulating key information related to its schedule and operations.

Closely tied to the Schedule and TrafficControlDepartment for operational management.

10. Schedule Class

Attributes:

scheduleID: int: Unique identifier for the train schedule.

trainNumber: String: Unique number identifying the train.

departureTime: String: The scheduled departure time of the train.

arrivalTime: String: The scheduled arrival time of the train.

Methods:

updateTime(): Updates the schedule in response to changes or delays.

Analysis:

Represents train schedules and their associated details.

Facilitates operational adjustments via the updateTime() method, ensuring accurate tracking.

11. Engine Class

Attributes:

engineID: int: Unique identifier for the engine.

location: String: Current location of the engine.

status: String: Operational status of the engine (e.g., active, under maintenance).

Methods:

trackEngine(): Tracks the engine's real-time status or location.

Analysis:

Encapsulates key details about the train's engine and its operational state.

Likely integrates with the MaintenanceDepartment for tracking and servicing.

12. Storage Class

Attributes:

Diesel_Barrels: int: Represents the quantity of diesel fuel available.

Steel_MetricTons: int: Tracks the quantity of steel resources.

Iron_MetricTons: int: Tracks the quantity of iron resources.

Methods:

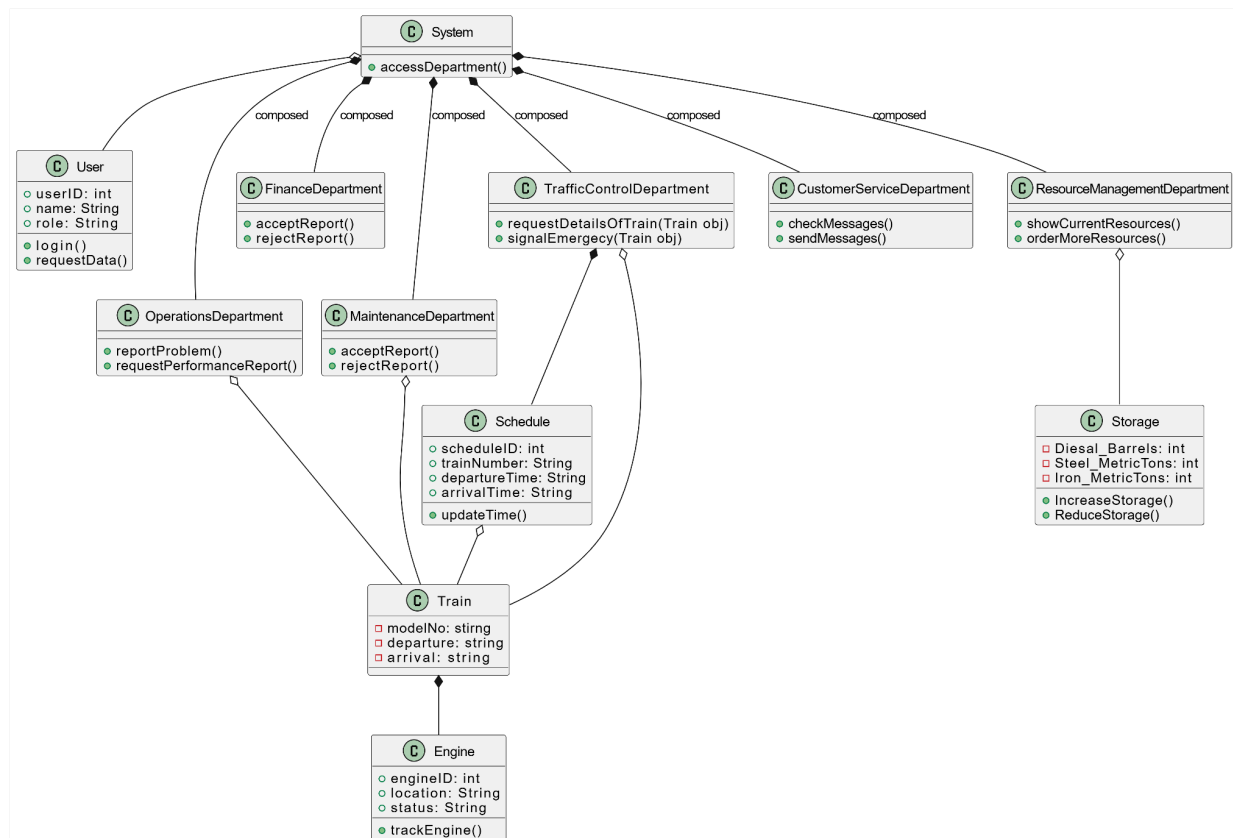
IncreaseStorage(): Adds resources to storage.

ReduceStorage(): Reduces resources as they are consumed.

Analysis:

Represents the physical resources required for operations and maintenance.

Closely connected to the ResourceManagementDepartment for replenishment and usage tracking.



▪ Operating environment

Supported operating systems:

Desktop: Windows, macOS, Linux.

Mobile: Android, iOS.

Others: ChromeOS

Supported browsers:

The website will work on the latest versions of

- Chromium-based browsers - e.g Google Chrome
- Gecko-based browsers - e.g Firefox
- EdgeHTML-based browsers - e.g Microsoft Edge (2015)

▪ Assumptions and dependencies

Assumptions:

- The website is accessible 24/7
- The hosting server of the website remains underloaded 24/7

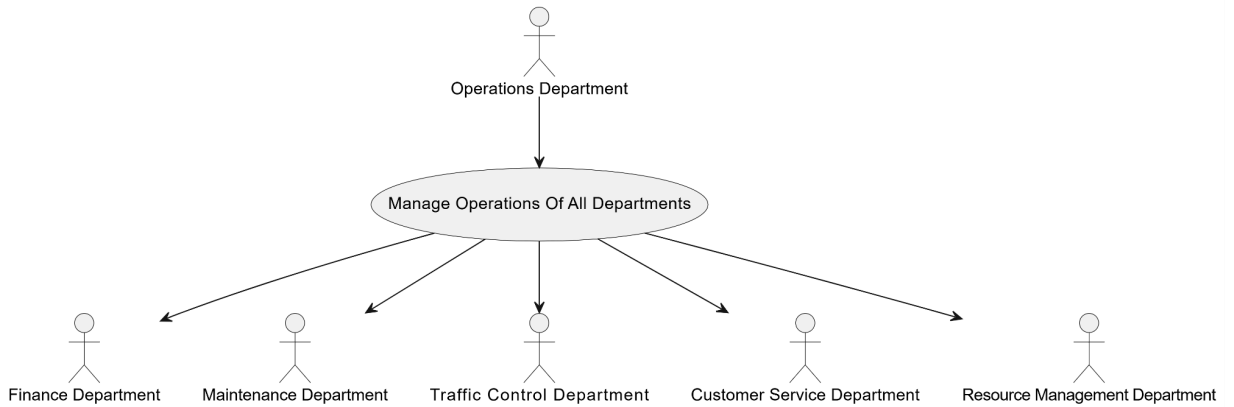
Dependencies:

- The website's operation depends on the permission of Ministry of Railways Pakistan

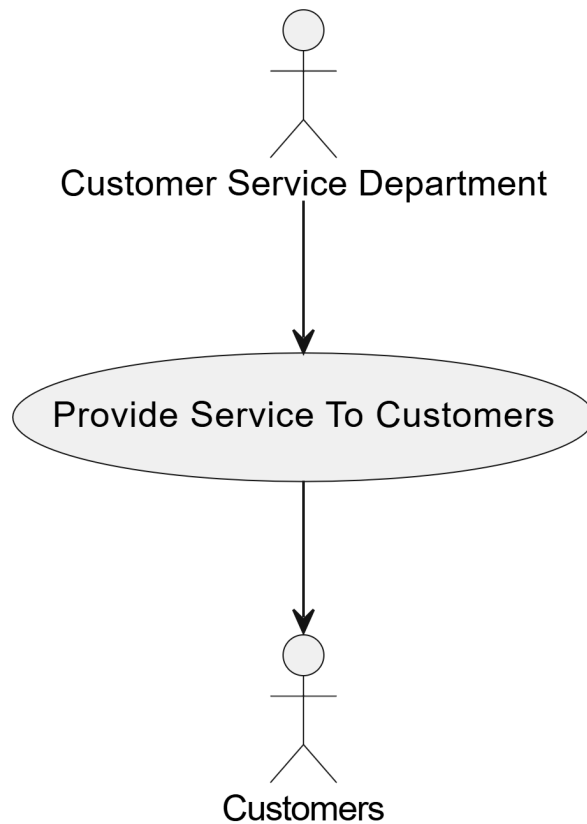
o Specific Requirements

- Functional requirements
- Non-functional requirements (e.g., performance, usability, security)
- External interface requirements (user, hardware, software, communication interfaces)

Use Cases



Description	The Operations Department manages the working of other departments, requests stats, reports notifications if a department is not functioning properly, and monitors the status of each department.
Actors	• Operations Department • Other departments
Basic Flow	1. The Operations Department logs into the central management portal. 2. Requests stats from other departments. 3. Monitors the status of each department. 4. Reports notifications if a department is not functioning properly.
Alternate Flow	If the portal login fails, the staff retries the login.
Pre-conditions	The operations department portal is working
Post-conditions	Department stats are accessed, notifications are reported if necessary, and the status of each department is monitored.

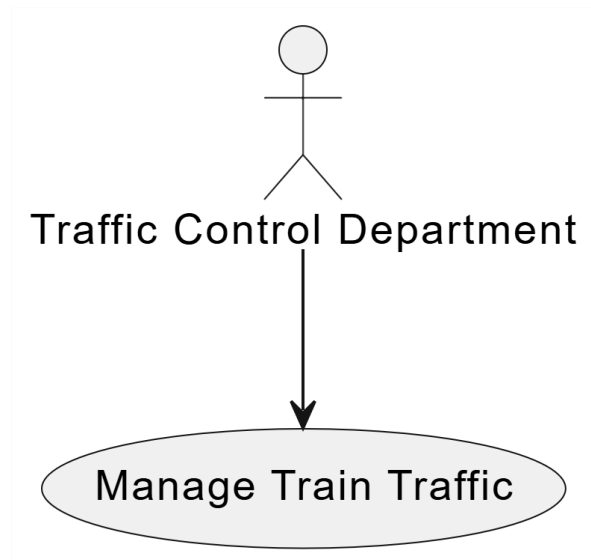


2.

Description	The Customer Service Department handles all customer complaints using calls and messages.
Actors	• Customer Service Department • Customers
Basic Flow	1. The customer contacts the Customer Service Department via call or message. 2. Customer Service Representative logs into the complaint management system. 3. The representative addresses and resolves the complaint. 4. The representative logs the resolution in the system.

Alternate Flow	If the customer service system is down, representatives note complaints manually and address them once the system is back up.login.
Pre-conditions	Complaint management system and communication lines (calls/messages) should be operational.
Post-conditions	Complaints are resolved, and the status is updated in the system.

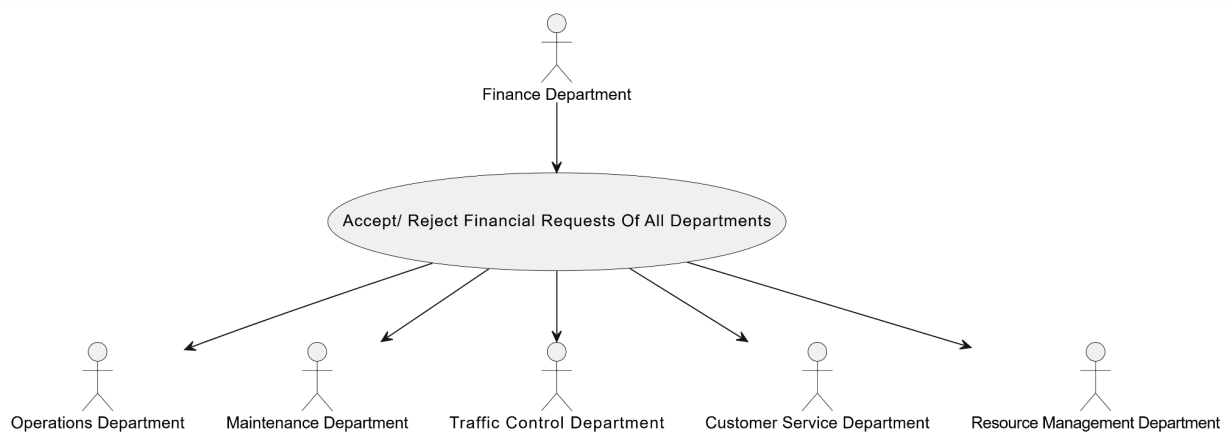
(3)



Description	Users in the Traffic Control Department access the portal to manage and monitor train traffic. They can view details such as the departure place, destination place, time of departure, time of arrival, and the status of the train (whether it has reached the destination, departed, or is on the way).
Actors	Traffic Control Department Staff Train Operators

Basic Flow	1. User logs into the Traffic Control Department portal. 2. User navigates to the train traffic management section. 3. User views the list of trains with details including departure place, destination place, time of departure, time of arrival, and status. 4. Users monitor and update the status of trains as needed.
Alternate Flow	1. If the user cannot access the portal, they should contact IT support. 2. If train details are not updated, the user should refresh the page or check back later.
Pre-conditions	Users must have access to the Traffic Control Department portal.
Post-conditions	Users can successfully view and manage train traffic information.

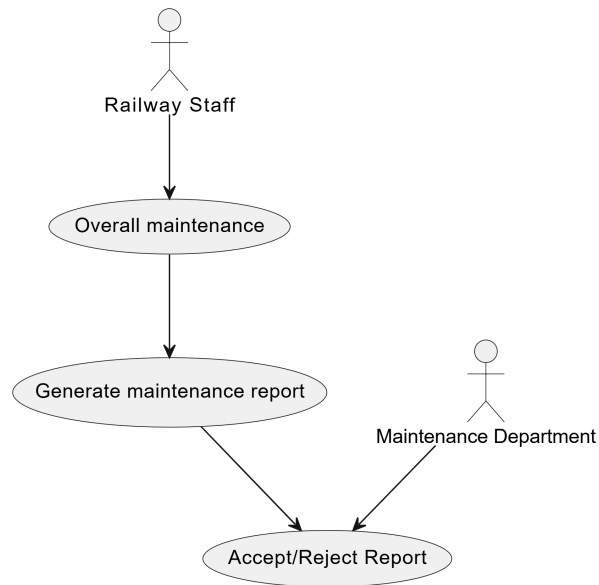
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Description	Users log into the Finance Department portal to view all financial requests made by other departments, including the amount of money
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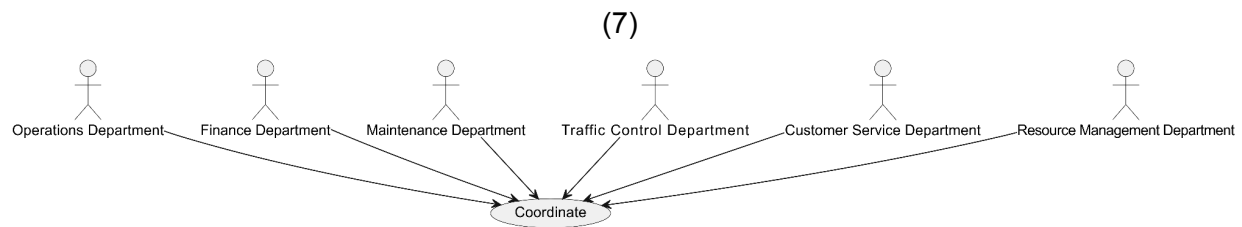
	requested and the reason for the request. The Finance Department has the authority to accept or reject these requests.
Actors	• Finance Department Staff • Users from other Departments (Operations, Maintenance, Traffic Control, Customer Service, Resource Management, etc.)
Basic Flow	1. Users log into the Finance Department portal. 2. Users submit financial requests, detailing the amount of money needed and the reason. 3. Finance Department staff review the requests. 4. The Finance Department decides to accept or reject the requests based on the provided details.
Alternate Flow	1. If the Finance Department portal is unavailable, users are notified and may need to submit requests via email or other means.
Pre-conditions	The Finance Department portal must be operational and accessible to users.
Post-conditions	Financial requests are either accepted or rejected, and the status is updated in the portal

(6)



Description	The Railway Staff performs overall maintenance of trains and uploads the maintenance report and statistics on the portal. The Maintenance Department receives the report, views the details, and has the authority to accept or reject the report.
Actors	• Railway Staff • Maintenance Department Staff
Basic Flow	1. Railway Staff performs the maintenance of trains. 2. Railway Staff generates and uploads the maintenance report and statistics on the portal. 3. Maintenance Department Staff logs into the portal and views the report.

	4. Maintenance Department Staff accepts or rejects the report based on the details provided.
Alternate Flow	1. If the portal is down, the Railway Staff may submit the report via an alternate method and the Maintenance Department reviews it once the portal is back up.
Pre-conditions	The maintenance portal must be operational and accessible to both Railway Staff and Maintenance Department Staff.
Post-conditions	The maintenance report is either accepted or rejected by the Maintenance Department.



Description	All the departments will coordinate online
Actors	• Operations department • Finance department • Maintenance department • Traffic control department • Customer service department • Resource management department • Operations department user logs in • Manage their own operations • All the departments coordinate
Basic Flow	1. Operations department user logs in 2. Manage their own operations 3. All the departments coordinate
Alternate Flow	Log in fails, employees logs in again

Pre-conditions	Website should be in working state
Post-conditions	Proper coordination ensures OR Coordination not ensured

Specific Requirements

- Functional requirements

- **R1** :The system should solve “delay in train departure and arrival”.
- **R2** :The system should maximize coordination between railways departments. through an online website.
- **R3** :The user data should be visible to all employees.
- **R4** :There should be real-time engine monitoring.

- Non-functional requirements

- **R5**: The system user age is from children to adults.
- **R6**: There are legal constraints from the Ministry of Railways.
- **R7**: System should be operational on less internet speed.
- **R8**: The app should respond to user actions (e.g., clicking a button, submitting a form) quickly.
- **R9**: The app should handle a lot of users without performance degradation.
- **R10**:The system should be operational most of the time.
- **R11**: Use authentication methods for all user accounts.
- **R12**: The website should use **HTTPS** protocol for the safety of user data.
- **R13**: The system should be restored quickly after a critical failure.

- External interface requirements

This section will cover User interface, Hardware interface, Software interface, Communication interface

1. User Interface (UI)

Describe how users will interact with the software.

- **Purpose:** The main goal of UI is to provide a user-friendly, decent, and sleek set of menus to make users able to use the website with confidence.
 - **Key Elements:**
 - The main elements include buttons, sliding menus, and background pictures, all aiming to produce focus towards establishing user-friendly experience.
 - Its demo screen will be shown in the Mockup section of the SRS.
 - **Standards/Guidelines:** The guidelines according to the Ministry of Railways are used to make UI
-

2. Hardware Interface

Describe how the software interacts with hardware components.

- **Purpose:** The purpose of this interface is to define the communication with hardware related to the external environment
 - **Supported Hardware:**
 - Printers
 - PCs
 - Mobile phones
-

3. Software Interface

- **Purpose:** The website will store data using JSON format
- **External Systems:**

- Specify APIs or libraries used.
- List software with which the system will communicate (e.g., databases, third-party services).
- **Data Exchange:**
 - Format of exchanged data (e.g., JSON, XML, CSV).
 - Method of data exchange (e.g., REST API, file transfer).

Example:

- The software integrates with an external payment gateway via REST API.
 - Data is exchanged in JSON format, secured with OAuth 2.0 authentication.
 - It also requires connection to MySQL for data storage.
-

4. Communications Interface

Describe how the software communicates with external systems, devices, or networks.

- **Purpose:** The website will use network protocol for communication with devices
- **Network Requirements:**
 - The protocol used is HTTPs

Stakeholder Feedback

- Interview

We conducted an interview of the stakeholder twice and the requirement engineer was different both times and after both requirement engineers. After the requirement engineers gathered the requirements of stakeholders they merged the requirements and made them concise and clear.

- Final Requirements Review

After the requirement engineers merged all the requirements by using interaction matrices and discussion they did another meeting with the stakeholder and showed him the final requirements.

- Addition of New Requirements

On the review of the final requirements, the stakeholder seemed unsatisfied with the non-functional requirements and wanted to add more of the non-functional requirements.

The new requirements requirements were:-

- The app should respond to user actions (e.g., clicking a button, or submitting a form) quickly.
- The app should handle a lot of users without performance degradation.
- The system should be operational most of the time.
- Use authentication methods for all user accounts.
- The website should use **HTTPS** protocol for the safety of user data.
- The system should be restored quickly after a critical failure.

- Satisfaction of Stakeholders

Upon the dissatisfaction of stakeholders regarding the NFRs, requirement engineers added new requirements requested by the stakeholders and stakeholders seemed satisfied with that.

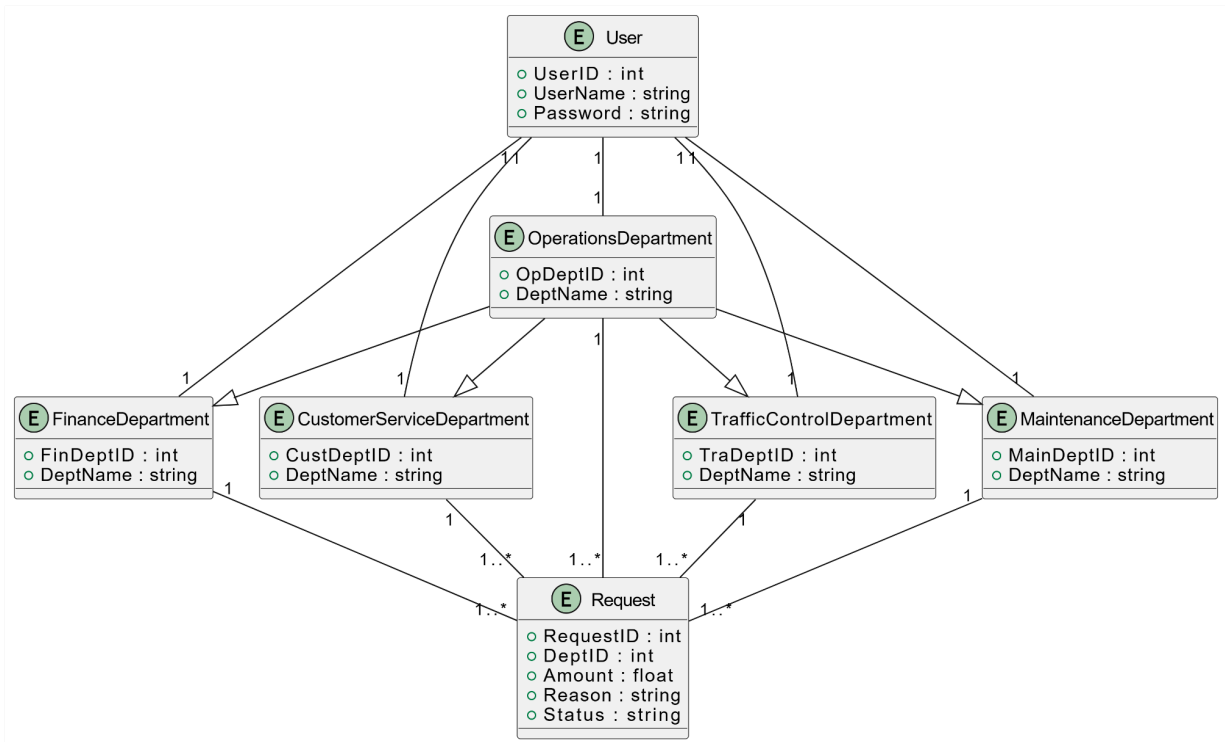
Technical Details

- NFR Metrics:-

Property	Measure
Scalability	The system should be able to handle 10,000 users at a time.

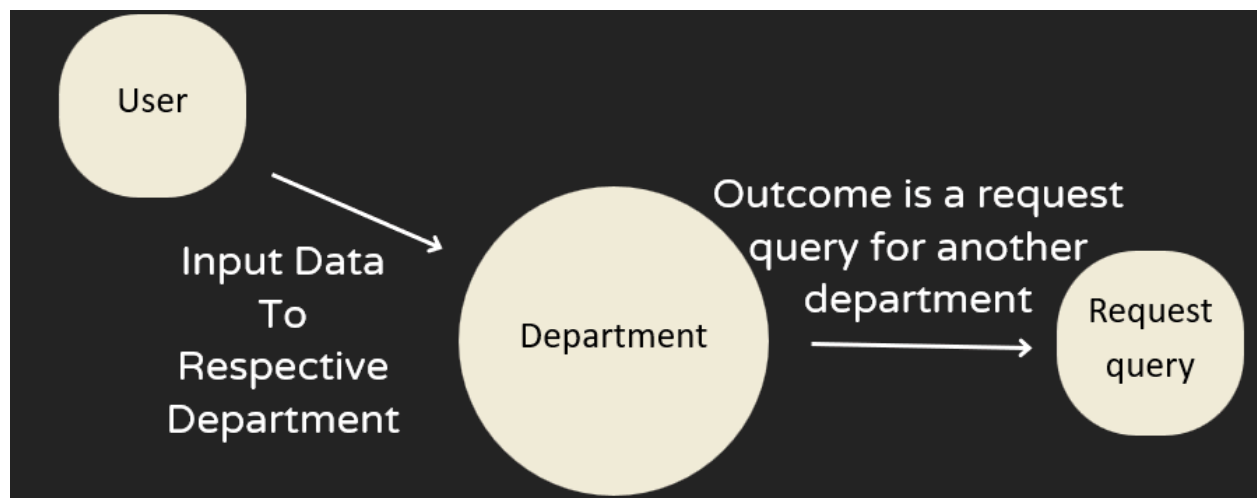
Security	System should use HTTPS protocol and 2FA authentication to give access to the system to any user.
Legal	System should only be accessed by people older than 13 and it should also follow all the governmental policies and does not violate any rules and regulations imposed by the government .
Robustness	System should be able to be restored within 2 hours after a critical failure.
Performance	System should take average time less than 10s and greater than 5s to respond to any command given by the user.
Reliability	System should be available 99.9% of the whole day.
Internet	System should have at least the internet connection 5mbps to operate correctly.

Entity relationship diagram:



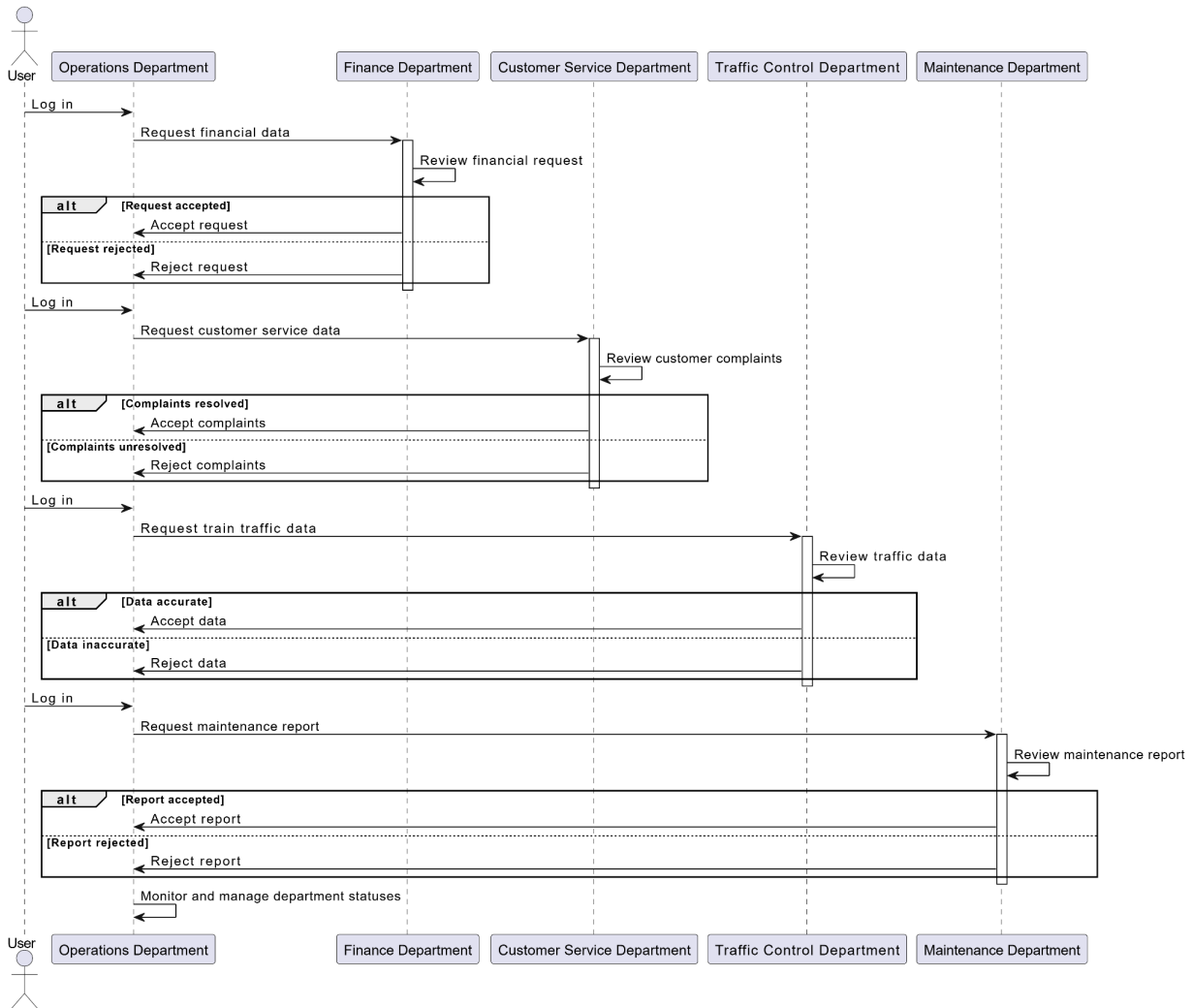
Dataflow diagram level 0:

The following diagram illustrates the conversion of data from one form to another throughout the website on an abstract level.



Sequence diagram:

The following diagram illustrates sequence of operations of operation department



Requirements Traceability Table

Project Goals

- **G1:** Project can solve the problem of the lack of coordination between different railway departments.
- **G2:** Project is scalable and can support multiple users
- **G3:** Project keeps the user data secure and follow government guidelines
- **G4:** Project performs the tasks efficiently.

Matrix

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13
G1	✓	✓	✓	✓									
G2									✓				
G3					✓	✓					✓	✓	
G4							✓	✓		✓			✓

Mockup

Please click on the following link to view the mockup of the Pakistan Railways website
<https://app.visily.ai/projects/7e81bcab-913e-4303-b520-6f6b3b21113f/boards/1406726>
